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DATA TRANSMISSION FROM EDGE TO CLOUD USING EDGE-BASED BIDDING

Abstract

A remote compute device stores a data collection policy. A processor receives a data request from an application and determines a plurality of data sets within the data request. The processor provides a data bid request to multiple edge compute devices. The processor receives multiple bidding values from the edge compute devices. A different one of the bidding values corresponds to a different one of the edge compute devices. Based on the multiple bidding values and the data collection policy, the processor determines a target edge compute device. The processor provides a data request to the target edge compute device. The processor receives data from the target edge compute device and provide the data to the application.

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Background/Summary

FIELD OF THE DISCLOSURE

[0001] The present disclosure generally relates to information handling systems, and more particularly relates to data transmission from an edge device to a cloud device using edge-based bidding.

BACKGROUND

[0002] As the value and use of information continues to increase, individuals and businesses seek additional ways to process and store information. One option is an information handling system. An information handling system generally processes, compiles, stores, or communicates information or data for business, personal, or other purposes. Technology and information handling needs and requirements can vary between different applications. Thus, information handling systems can also vary regarding what information is handled, how the information is handled, how much information is processed, stored, or communicated, and how quickly and efficiently the information can be processed, stored, or communicated. The variations in information handling systems allow information handling systems to be general or configured for a specific user or specific use such as financial transaction processing, airline reservations, enterprise data storage, or global communications. In addition, information handling systems can include a variety of hardware and software resources that can be configured to process, store, and communicate information and can include one or more computer systems, graphics interface systems, data storage systems, networking systems, and mobile communication systems. Information handling systems can also implement various virtualized architectures. Data and voice communications among information handling systems may be via networks that are wired, wireless, or some combination.

SUMMARY

[0003] A remote compute device may store a data collection policy. A processor may receive a data request from an application and determine a plurality of data sets within the data request. The processor may provide a data bid request to multiple edge compute devices. The processor may receive multiple bidding values from the edge compute devices. A different one of the bidding values corresponds to a different one of the edge compute devices. Based on the multiple bidding values and the data collection policy, the processor may determine a target edge compute device. The processor may provide a data request to the target edge compute device. The processor may receive data from the target edge compute device and provide the data to the application.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] It will be appreciated that for simplicity and clarity of illustration, elements illustrated in the Figures are not necessarily drawn to scale. For example, the dimensions of some elements may be exaggerated relative to other elements. Embodiments incorporating teachings of the present disclosure are shown and described with respect to the drawings herein, in which:

[0005] FIG. **1** is a block diagram of a portion of a system including multiple information handling systems, a remote compute device, a backend server, and multiple edge compute devices according to at least one embodiment of the present disclosure;

[0006] FIG. **2** is a flow diagram of a method for performing data transmission from an edge device to a remote compute device based on edge-based bidding according to at least one embodiment of the present disclosure; and

[0007] FIG. **3** is a block diagram of a general information handling system according to an embodiment of the present disclosure.

[0008] The use of the same reference symbols in different drawings indicates similar or identical items.

DETAILED DESCRIPTION OF THE DRAWINGS

[0009] The following description in combination with the Figures is provided to assist in understanding the teachings disclosed herein. The description is focused on specific implementations and embodiments of the teachings and is provided to assist in describing the teachings. This focus should not be interpreted as a limitation on the scope or applicability of the teachings.

[0010] FIG. **1** illustrates a system **100** including multiple information handling systems **102** and **104**, a remote compute device **106**, backend server **108**, and edge compute devices **110**, **112**, **114**, **116**, **118**, **120**, and **122** (**110-122**) according to at least one embodiment of the present disclosure. For purposes of this disclosure, an information handling system can include any instrumentality or aggregate of instrumentalities operable to compute, calculate, determine, classify, process, transmit, receive, retrieve, originate, switch, store, display, communicate, manifest, detect, record, reproduce, handle, or utilize any form of information, intelligence, or data for business, scientific, control, or other purposes. For example, an information handling system may be a personal computer (such as a desktop or laptop), tablet computer, mobile device (such as a personal digital assistant (PDA) or smart phone), server (such as a blade server or rack server), a network storage device, or any other suitable device and may vary in size, shape, performance, functionality, and price. The information handling system may include random access memory (RAM), one or more processing resources such as a central processing unit (CPU) or hardware or software control logic, ROM, and/or other types of nonvolatile memory. Additional components of the information handling system may include one or more disk drives, one or more network ports for communicating with external devices as well as various input and output (I/O) devices, such as a keyboard, a mouse, touchscreen and/or a video display. The information handling system may also include one or more buses operable to transmit communications between the various hardware components.

[0011] Information handling system **102** includes a processor **130**, a memory **132**, and a control plane **134**. Information handling system **104** includes a processor **140**, a memory **142**, and a control plane **144**. Remote compute device **106** includes a processor **150** and a memory **152**. Processor **150** may execute any suitable application or service, such as telemetry service **154**. Backend server **108** includes a processor **160** and a memory **162**. In an example, memory **162** may store any suitable data for system **100** including, but not limited to, a data collection policy, such as an information technology decision maker (ITDM) policy **164**. In an example, ITDM policy **164** may include any suitable criteria for edge-bidding in edge compute devices **110-122** including, but not limited to, a best reliability, a best data transmission cost, best carbon footprint, and fastest edge compute device.

[0012] Edge compute device **110** includes a processor **170**, a memory **172**, and a control plane **174**. Each of the remaining edge compute devices **112-122** includes a processor, a memory, and a control plane. For clarity and brevity each of edge compute devices **112-122** are illustrated with only a control plane **174**. System **100** may include any suitable number of information handling systems, edge compute devices, remote compute devices, and backend servers without varying from the scope of this disclosure. Each of information handling systems **102** and **104**, remote compute device **106**, backend server **108**, and edge compute devices **110-122** may include any suitable components without varying from the scope of this disclosure.

[0013] System **100** may be any suitable infrastructure for transferring data between information handling systems edge-cloud infrastructure. In certain examples, edge compute devices **110-122** may be located in different geographic regions, and access to the edge compute devices by information handling systems **102** and **104** may varying based on the location of the geographic regions and the location of the information handling systems. System **100** may include any suitable number of communication channels connected through different networks to provide communication among information handling systems **102** and **104**, remote compute device **106**, backend server **108**, and edge compute devices **110-122**. For example, information handling systems **102** and **104** may communicate with to receive/provide data from/to backend server **108** and any and all of edge compute devices **110-122**. Remote compute device **106** may communicate with to receive/provide data from/to backend server **108** and any and all of edge compute devices **110-112**. Edge compute devices **110-112** may communicate with to receive/provide data from/to each other, information handling systems **102** and **104**, remote compute device **106**, and backend server **108**.

[0014] During operation of the devices of system **100**, edge compute devices **110-122** may receive data from the end-point devices, such as information handling systems **102** and **104**, and perform certain data analytics on the received data. In certain examples, larger deep artificial intelligence-based (AI-based) data analysis may be performed at the cloud level, such as by remote compute device **106**. This deep AI-based analysis may be performed by remote compute device **106** based on the remote compute device having access to larger sets of data. While remote compute device **106** is performing the deep AI-based analysis, the remote compute device may request data sets from edge compute devices **110-122**. In certain examples, remote compute device **106** may request the data at any suitable points in time, such as periodically, on-demand, or the like. The data set requests may be performed to enable remote compute device **106** to receive data for processing or performing data analysis within remote compute device **106**. In certain examples, multiple edge compute devices **110-122** may hold or store the same data in the corresponding memory, such as memory **172** of edge compute device **110**. In current edge-cloud infrastructure systems, there is no optimized scheme available for a remote compute device to request data from a proper or correct edge compute device. AI-based data analysis performed by processor **150** may be improved by an edge-based bidding process enabling remote compute device **106** to request the data sets from the right edge compute device **110-122**.

[0015] In certain examples, edge compute devices **110-122** may be located in different geographic regions **180**, **182**, and **184**. These regions **180**, **182**, and **184** may be located across a state, a country, or the world. As illustrated in FIG. **1**, edge compute devices **110**, **112**, and **114** may be located in region **180**, edge compute devices **116** and **118** may be located in region **182**, and edge compute devices **120** and **122** may be located in region **184**. In certain examples, remote compute device **106** and backend server **108** may be located within any suitable region, such as region **180**. Information handling systems **102** and **104** may be located in any suitable region of system **100**. In an example, regions **180**, **182**, and **184** may include any suitable number of edge compute devices without varying from the scope of this disclosure.

[0016] In an example, processor **160** or any other suitable component within backend server **108** may communicate with remote compute device **106** and each control plane **174** of edge compute devices **110** to provide ITDM policy **164** to the remote compute device and the edge compute devices. In response to receiving ITDM policy **164**, remote compute device **106** may store the ITDM policy in memory **152** for later use. Similarly, each of edge compute devices **110-122** may store ITDM policy **164** in respective memories, such as memory **172** of edge compute device **110**, for later use.

[0017] In an example, remote compute device **106** may provide a data bid request to each of edge compute devices **110-122**. Based on reception of the data bid request, each of edge compute devices **110-122** may determine a respective bid value. In certain examples, each of edge compute device **110-122** may determine a bid value in substantially the same manner. For clarity and brevity, the determination of a bid value will be described with respect to a single edge compute device, such as edge compute device **110**. In an example, processor **170** may perform one or more suitable operations to determine the bid value for edge compute device **110**. For example, processor **170** may determine workload load, subsystem health, data transmission cost, carbon intensity, and other factors of edge compute device **110** to determine the bidding value. These factors may provide information about conditions within edge compute device **110** that may affect the ability of the edge compute device to provide telemetry and other data from information handling system **102** to remote compute device **106**.

[0018] In certain examples, the bidding value may be a numerical value based on an overall ability of edge compute device **110** to provide the telemetry and other data. For example, processor **170** may utilize ITDM policy **164** stored in memory **172** to determine an overall bidding value based on the combination of factors stated above. In an example, the bidding value may be any particular value in a range of bidding values. In one example, as the bidding value increases the ability of edge compute device **110** to provide the telemetry and other data may also increase. In an alternative example, as the bidding value decreases the ability of edge compute device **110** to provide telemetry and other data may increase. In an example, the bidding value may be a set of values and each value may be based on the different factors determined by processor **170**. For example, the bidding value may include a value for workload load, a value for subsystem health, a value for data transmission cost, a value for carbon intensity, and values for other factors of edge compute device **110**.

[0019] In an example, ITDM **164** or any other suitable policy may set a level of granularity for the bidding values. For example, the larger the range of possible bidding values, the larger the granularity. In an example, as the granularity increases for the bidding values, the less likely that multiple edge compute devices **110-122** may provide the same bidding value to remote compute device **106**. In this situation, as the number of edge compute devices **110-122** increases within system **100**, the granularity of the range of bidding values should increase to prevent the edge compute devices from providing the same bidding value to remote compute device **106**. In an example, the bidding value for each of the factors may be any particular value in a range of bidding values. In certain examples, the range for each of the factors may be the same or the range for each of the factors may vary among the factors. In an example, if a workload on processor **170** is above a threshold level, processor **170** may determine that the bidding value for edge compute device **110** should be below a particular value.

[0020] In response to receiving the bidding values from each of edge compute devices **110-122**, processor **150** in remote compute device **106** may perform any suitable operations to determine a target edge compute device to provide the data request. In an example, processor **150** may utilize ITDM policy **164** stored in memory **152** to determine the target edge compute device. For example, processor **170** may utilize the bidding values from each of edge compute devices **110-122** and criteria in IDTM policy **164** to determine a target edge compute device. IDTM policy **164** may define criteria for processor **170** to determine a target edge compute device based on one or more of a best reliability, a best data transmission cost, best carbon footprint, a fastest edge compute device, or the like.

[0021] Based on processor **150** determining a target edge compute device, such as edge compute device **110**, the processor may execute telemetry service **154** may communicate over a network to provide the data request to target edge compute device **110**. In an example, a data lane may be created between remote compute device **107** and edge compute device **110** and this data lane may be utilized to provide the requested data, such as platform-level hardware data and behavior data, for information handling system **102** to remote compute device **106**. In response to receiving the requested data from edge compute device **110**, processor **150**, via telemetry service **154**, may automatically provide the received data to one or more consumer applications in remote compute device **106**. Thus, remote compute device **106** and edge compute devices **110-122** may utilize a bidding process to retrieve data for information handling system **102** or **104** from the proper edge compute device of edge compute devices **110-122**.

[0022] FIG. **2** is a flow diagram of a method **200** for performing data transmission from an edge device to a remote compute device based on edge-based bidding according to at least one embodiment of the present disclosure, starting at block **202**. It will be readily appreciated that not every method step set forth in this flow diagram is always necessary, and that certain steps of the methods may be combined, performed simultaneously, in a different order, or perhaps omitted, without varying from the scope of the disclosure. FIG. **2** may be employed in whole, or in part, processor **150** of remote compute device **104**, processor **160** of backend server **108**, and processors of edge compute devices **110-122**, such as processor **170** of edge device **110**, in FIG. **1**, or any other type of controller, device, module, processor, or any combination thereof, operable to employ all, or portions of, the method of FIG. **2**.

[0023] At block **204**, an analytics process is started. In an example, the analytics process may be started in a remote compute device. In certain examples, the remote compute device may include one or more application, such as AI-based applications to perform the analytics process or operations on data from one or more information handling systems. The data from the information handling system may be any suitable data, such as platform-level data and behavior data. During the analytics operations, the applications in the remote compute device may need additional data.

[0024] At block **206**, a list of data sets for the analytics process is created. In an example, the list of data sets may include any suitable information to identify the data sets needed by the applications to continue performing the analytics process or operations. A processor of the remote compute device may need different data sets based on the analytics operations being performed by the applications in the remote compute device. In certain examples, the list of data sets may be stored in a memory of the remote compute device.

[0025] At block **208**, all available edge compute devices are enumerated. In certain examples, multiple edge compute devices in an IT infrastructure system may include the same data from one or more information handling systems of the system. In an example, remote compute devices may determine the available edge compute devices by determining which edge compute devices in an IT infrastructure system are currently powered on and actively communicating with other devices of the system.

[0026] At block **210**, data bid requests are broadcasted to all available edge compute devices. In an example, remote compute device may provide the data bid requests to all available edge compute devices by one or more network connections and the control planes of the edge compute devices. At block **212**, data bid request responses are received from all of the available edge compute devices. In certain examples, the edge compute devices may determine a bidding value based on workload load, subsystem health, data transmission cost, carbon intensity, and other factors of edge compute device. These factors may provide information about conditions within the edge compute device that may affect the ability of the edge compute device to provide telemetry and other data from the information handling system to the remote compute device.

[0027] In certain examples, the bidding value may be a numerical value based on an overall ability of the edge compute device to provide the telemetry and other data. For example, the edge compute device may utilize an ITDM policy to determine an overall bidding value based on the combination of factors stated above. In an example, the bidding value may be a set of values and each value may be based on the different factors determined by a processor of the edge compute device.

[0028] At block **214**, a target edge device is determined. In an example, a processor of the remote compute device may perform any suitable operations to determine the target edge compute device. For example, the remote compute device may determine the target edge compute device based on the received bidding values from the edge compute devices. The remote compute device may also utilize the ITDM policy for the IT infrastructure system to determine the target edge compute device.

[0029] At block **216**, the data request is provided to the target edge compute device. In an example, the remote compute device may execute telemetry service may communicate over a network to provide the data request to the target edge compute device. In certain examples, a data lane may be created between the remote compute device and the edge compute device and this data lane may be utilized to provide the requested data, such as platform-level hardware data and behavior data, to the target remote compute device.

[0030] At block **218**, collected data is received from the target edge compute device. At block **220**, the analytics process is performed and the flow ends at block **222**. In response to receiving the requested data from the edge compute device, the remote compute device may automatically provide the received data to one or more consumer applications. In an example, the applications may performed AI-based data analytics operations on the data received from the target edge compute device.

[0031] FIG. **3** shows a generalized embodiment of an information handling system **300** according to an embodiment of the present disclosure. Information handling system **300** may be substantially similar to information handling system **102**, remote compute device **104**, backend server **108**, and edge device **170** of FIG. **1**. For purpose of this disclosure an information handling system can include any instrumentality or aggregate of instrumentalities operable to compute, classify, process, transmit, receive, retrieve, originate, switch, store, display, manifest, detect, record, reproduce, handle, or utilize any form of information, intelligence, or data for business, scientific, control, entertainment, or other purposes. For example, information handling system **300** can be a personal computer, a laptop computer, a smart phone, a tablet device or other consumer electronic device, a network server, a network storage device, a switch router or other network communication device, or any other suitable device and may vary in size, shape, performance, functionality, and price. Further, information handling system **300** can include processing resources for executing machine-executable code, such as a central processing unit (CPU), a programmable logic array (PLA), an embedded device such as a System-on-a-Chip (SoC), or other control logic hardware. Information handling system **300** can also include one or more computer-readable medium for storing machine-executable code, such as software or data. Additional components of information handling system **300** can include one or more storage devices that can store machine-executable code, one or more communications ports for communicating with external devices, and various input and output (I/O) devices, such as a keyboard, a mouse, and a video display. Information handling system **300** can also include one or more buses operable to transmit information between the various hardware components.

[0032] Information handling system **300** can include devices or modules that embody one or more of the devices or modules described below and operates to perform one or more of the methods described below. Information handling system **300** includes a processors **302** and **304**, an input/output (I/O) interface **310**, memories **320** and **325**, a graphics interface **330**, a basic input and output system/universal extensible firmware interface (BIOS/UEFI) module **340**, a disk controller **350**, a hard disk drive (HDD) **354**, an optical disk drive (ODD) **356**, a disk emulator **360** connected to an external solid state drive (SSD) **362**, an I/O bridge **370**, one or more add-on resources **374**, a trusted platform module (TPM) **376**, a network interface **380**, a management device **390**, and a power supply **395**. Processors **302** and **304**, I/O interface **310**, memory **320**, graphics interface **330**, BIOS/UEFI module **340**, disk controller **350**, HDD **354**, ODD **356**, disk emulator **360**, SSD **362**, I/O bridge **370**, add-on resources **374**, TPM **376**, and network interface **380** operate together to provide a host environment of information handling system **300** that operates to provide the data processing functionality of the information handling system. The host environment operates to execute machine-executable code, including platform BIOS/UEFI code, device firmware, operating system code, applications, programs, and the like, to perform the data processing tasks associated with information handling system **300**.

[0033] In the host environment, processor **302** is connected to I/O interface **310** via processor interface **306**, and processor **304** is connected to the I/O interface via processor interface **308**. Memory **320** is connected to processor **302** via a memory interface **322**. Memory **325** is connected to processor **304** via a memory interface **327**. Graphics interface **330** is connected to I/O interface **310** via a graphics interface **332** and provides a video display output **336** to a video display **334**. In a particular embodiment, information handling system **300** includes separate memories that are dedicated to each of processors **302** and **304** via separate memory interfaces. An example of memories **320** and **330** include random access memory (RAM) such as static RAM (SRAM), dynamic RAM (DRAM), non-volatile RAM (NV-RAM), or the like, read only memory (ROM), another type of memory, or a combination thereof.

[0034] BIOS/UEFI module **340**, disk controller **350**, and I/O bridge **370** are connected to I/O interface **310** via an I/O channel **312**. An example of I/O channel **312** includes a Peripheral Component Interconnect (PCI) interface, a PCI-Extended (PCI-X) interface, a high-speed PCI-Express (PCIe) interface, another industry standard or proprietary communication interface, or a combination thereof. I/O interface **310** can also include one or more other I/O interfaces, including an Industry Standard Architecture (ISA) interface, a Small Computer Serial Interface (SCSI) interface, an Inter-Integrated Circuit (I.sup.2C) interface, a System Packet Interface (SPI), a Universal Serial Bus (USB), another interface, or a combination thereof. BIOS/UEFI module **340** includes BIOS/UEFI code operable to detect resources within information handling system **300**, to provide drivers for the resources, initialize the resources, and access the resources. BIOS/UEFI module **340** includes code that operates to detect resources within information handling system **300**, to provide drivers for the resources, to initialize the resources, and to access the resources.

[0035] Disk controller **350** includes a disk interface **352** that connects the disk controller to HDD **354**, to ODD **356**, and to disk emulator **360**. An example of disk interface **352** includes an Integrated Drive Electronics (IDE) interface, an Advanced Technology Attachment (ATA) such as a parallel ATA (PATA) interface or a serial ATA (SATA) interface, a SCSI interface, a USB interface, a proprietary interface, or a combination thereof. Disk emulator **360** permits SSD **364** to be connected to information handling system **300** via an external interface **362**. An example of external interface **362** includes a USB interface, an IEEE 4394 (Firewire) interface, a proprietary interface, or a combination thereof. Alternatively, solid-state drive **364** can be disposed within information handling system **300**.

[0036] I/O bridge **370** includes a peripheral interface **372** that connects the I/O bridge to add-on resource **374**, to TPM **376**, and to network interface **380**. Peripheral interface **372** can be the same type of interface as I/O channel **312** or can be a different type of interface. As such, I/O bridge **370** extends the capacity of I/O channel **312** when peripheral interface **372** and the I/O channel are of the same type, and the I/O bridge translates information from a format suitable to the I/O channel to a format suitable to the peripheral channel **372** when they are of a different type. Add-on resource **374** can include a data storage system, an additional graphics interface, a network interface card (NIC), a sound/video processing card, another add-on resource, or a combination thereof. Add-on resource **374** can be on a main circuit board, on separate circuit board or add-in card disposed within information handling system **300**, a device that is external to the information handling system, or a combination thereof.

[0037] Network interface **380** represents a NIC disposed within information handling system **300**, on a main circuit board of the information handling system, integrated onto another component such as I/O interface **310**, in another suitable location, or a combination thereof. Network interface device **380** includes network channels **382** and **384** that provide interfaces to devices that are external to information handling system **300**. In a particular embodiment, network channels **382** and **384** are of a different type than peripheral channel **372** and network interface **380** translates information from a format suitable to the peripheral channel to a format suitable to external devices. An example of network channels **382** and **384** includes InfiniBand channels, Fibre Channel channels, Gigabit Ethernet channels, proprietary channel architectures, or a combination thereof. Network channels **382** and **384** can be connected to external network resources (not illustrated). The network resource can include another information handling system, a data storage system, another network, a grid management system, another suitable resource, or a combination thereof.

[0038] Management device **390** represents one or more processing devices, such as a dedicated baseboard management controller (BMC) System-on-a-Chip (SoC) device, one or more associated memory devices, one or more network interface devices, a complex programmable logic device

(CPLD), and the like, which operate together to provide the management environment for information handling system **300**. In particular, management device **390** is connected to various components of the host environment via various internal communication interfaces, such as a Low Pin Count (LPC) interface, an Inter-Integrated-Circuit (I2C) interface, a PCIe interface, or the like, to provide an out-of-band (OOB) mechanism to retrieve information related to the operation of the host environment, to provide BIOS/UEFI or system firmware updates, to manage non-processing components of information handling system **300**, such as system cooling fans and power supplies. Management device **390** can include a network connection to an external management system, and the management device can communicate with the management system to report status information for information handling system **300**, to receive BIOS/UEFI or system firmware updates, or to perform other task for managing and controlling the operation of information handling system **300**.
[0039] Management device **390** can operate off of a separate power plane from the components of the host environment so that the management device receives power to manage information handling system **300** when the information handling system is otherwise shut down. An example of management device **390** include a commercially available BMC product or other device that operates in accordance with an Intelligent Platform Management Initiative (IPMI) specification, a Web Services Management (WSMan) interface, a Redfish Application Programming Interface (API), another Distributed Management Task Force (DMTF), or other management standard, and can include an Integrated Dell Remote Access Controller (iDRAC), an Embedded Controller (EC), or the like. Management device **390** may further include associated memory devices, logic devices, security devices, or the like, as needed, or desired.
[0040] Although only a few exemplary embodiments have been described in detail herein, those skilled in the art will readily appreciate that many modifications are possible in the exemplary embodiments without materially departing from the novel teachings and advantages of the embodiments of the present disclosure. Accordingly, all such modifications are intended to be included within the scope of the embodiments of the present disclosure as defined in the following claims. In the claims, means-plus-function clauses are intended to cover the structures described herein as performing the recited function and not only structural equivalents, but also equivalent structures.

Claims

1. A remote compute device comprising: a memory to store a data collection policy; and a processor to communicate with the memory, the processor to: receive a data request from an application; determine a data set within the data request; provide a data bid request to multiple edge compute devices; receive multiple bidding values from the edge compute devices, wherein a different one of the bidding values corresponds to a different one of the edge compute devices; based on the multiple bidding values and the data collection policy, determine a target edge compute device; provide a data request for the data set to the target edge compute device; receive the data set from the target edge compute device; and provide the data set to the application.
2. The remote compute device of claim 1, wherein the processor further to: receive the data collection policy from a backend server, the backend server is in communication with the remote compute device, the multiple edge compute devices, and multiple information handling systems.
3. The remote compute device of claim 1, wherein prior to the data request being provided to the target edge compute device, the processor further to: create a data lane between the remote compute device and the target edge compute device, wherein the data is received from the target edge compute device over the data lane.
4. The remote compute device of claim 1, wherein each of the multiple edge compute devices include the data set.
5. The remote compute device of claim 1, wherein the data collection policy includes a level of

granularity for the bidding values.
6. The remote compute device of claim 1, wherein the each different one of the bidding values are based on workload load, subsystem health, data transmission cost, and carbon intensity of a corresponding different one of the edge compute devices.
7. The remote compute device of claim 1, wherein the data collection policy includes a reliability, a data transmission cost, carbon footprint, and speed for an edge compute device.
8. The remote compute device of claim 1, wherein the application is an artificial intelligence based data analysis application.
9. A method comprising: storing, by a processor of a remote compute device, a data collection policy in a memory; receiving a data request from an application; determining a plurality of data sets within the data request; providing a data bid request to multiple edge compute devices; receiving multiple bidding values from the edge compute devices, wherein a different one of the bidding values corresponds to a different one of the edge compute devices; based on the multiple bidding values and the data collection policy, determining a target edge compute device; providing a data request for the data set to the target edge compute device; receiving, by the processor, the data set from the target edge compute device; and providing the data set to the application.
10. The method of claim 9, further comprising receiving the data collection policy from a backend server, the backend server is in communication with the remote compute device, the multiple edge compute devices, and multiple information handling systems.
11. The method of claim 9, wherein prior to the data request being provided to the target edge compute device, the processor further to: creating a data lane between the remote compute device and the target edge compute device, wherein the data is received from the target edge compute device over the data lane.
12. The method of claim 9, wherein each of the multiple edge compute devices include the data set.
13. The method of claim 9, wherein the data collection policy includes a level of granularity for the bidding values.
14. The method of claim 9, wherein the each different one of the bidding values are based on workload load, subsystem health, data transmission cost, and carbon intensity of a corresponding different one of the edge compute devices.
15. The method of claim 9, wherein the data collection policy includes a reliability, a data transmission cost, carbon footprint, and speed for an edge compute device.
16. The method of claim 9, wherein the application is an artificial intelligence based data analysis application.
17. A remote compute device comprising: a memory to store a data collection policy, wherein the data collection policy is received from a backend server; and a processor to: receive a data request from an application; determine a data set within the data request; provide a data bid request to multiple edge compute devices; receive multiple bidding values from the edge compute devices, wherein a different one of the bidding values corresponds to a different one of the edge compute devices; based on the multiple bidding values and the data collection policy, determine a target edge compute device; provide a data request for the data set to the target edge compute device; create a data lane between the remote compute device and the target edge compute device; receive the data set from the target edge compute device over the data lane; and provide the data set to the application.
18. The remote compute device of claim 17, wherein each of the multiple edge compute devices include the data set.
19. The remote compute device of claim 17, wherein the each different one of the bidding values are based on workload load, subsystem health, data transmission cost, and carbon intensity of a corresponding different one of the edge compute devices.
20. The remote compute device of claim 17, wherein the data collection policy includes a level of granularity for the bidding values.